Private Equity in Public Pension Portfolios

FY2001-2025 | Independent descriptive research

5,061	235	2,961
CLEANED PLAN-YEARS	PUBLIC PENSION PLANS	MATCHED RETURN PAIRS

This project audits 5,061 plan-year records from 235 US public pension plans and compares reported private-equity (PE) and public-equity returns. PE's pooled median annual return is 11.12% (N=2,991), versus 12.61% for public equity (N=4,769). On 2,961 matched plan-years, return correlation is 0.526, falling to 0.261 after year demeaning. Allocation-zero treatment materially changes the allocation median, while tail clipping leaves return medians unchanged. Findings are descriptive, not causal.

2. Objective and data source

Compare pension-level PE and public-equity performance, allocation patterns and reporting quality using the Public Plans Database API. The unit of observation is a plan and fiscal year; these are reported asset-class annual returns, not individual-fund cash flows.

Public source: https://publicplansdata.org/api/ . The saved snapshot is audited here; live API metadata and the extraction timestamp were not supplied or independently verified.

3. Data pipeline

Stage	Verified record count / scope
Saved raw API export	13,007
Rows removed: all investment fields missing	7,946
Final unique plan-year records	5,061
Plans / fiscal years	235 / 25 (FY2001-2025)
Nonempty duplicate rows / conflicting groups	0 / 0

The supplied extraction script validates requested API fields, parses numeric columns, removes empty investment records and checks conflicting duplicates before first-nonmissing aggregation. Reconstruction from the saved raw CSV agrees with the cleaned panel; all numeric summaries reconcile. The extension reads originals without overwriting them.

4. Methodology

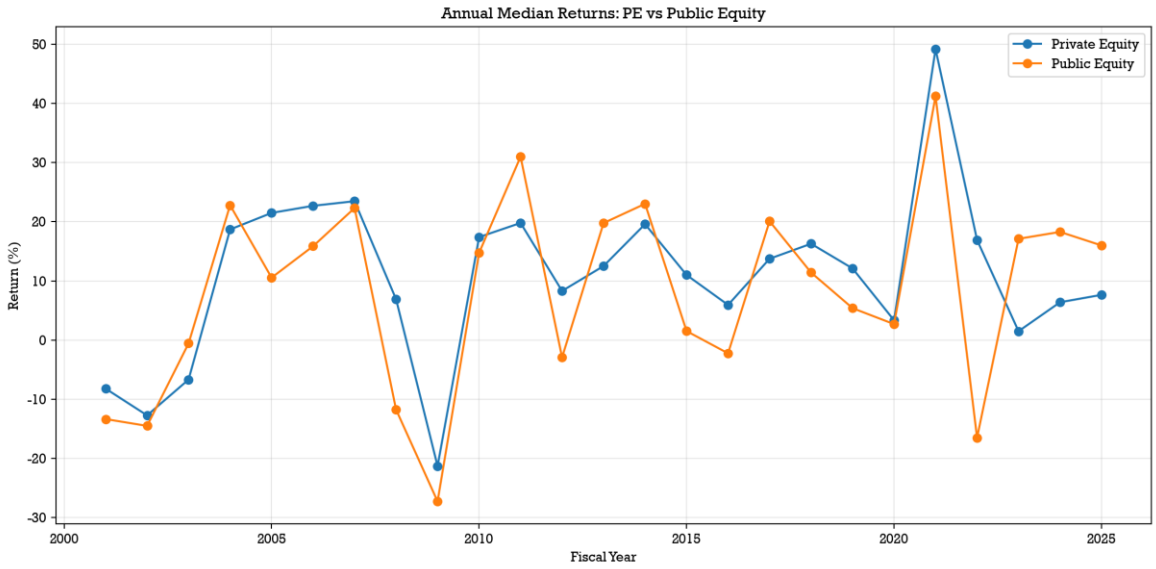
Use equally weighted available plan-years for pooled statistics and exact plan-year pairs for asset comparisons. Report N for each result. Treat zero allocation with no corresponding return as missing in an added column; retain every original row. Allocation bands are right-closed. Clip each asset's pooled returns at its empirical 1st/99th percentiles; leave missing values missing. For correlations, remove each year's mean on the same pair-complete sample. Compare annual median PE returns with current and fiscal-year-lagged equity medians. Fit year-effect regressions with plan-clustered standard errors where available.

Code excerpt | Conditional allocation recoding; d is the cleaned panel.

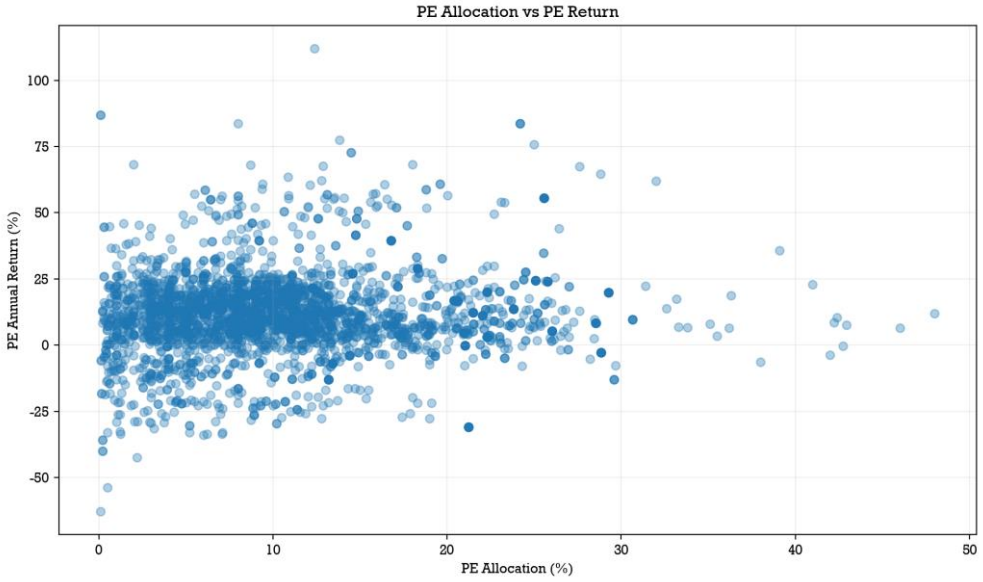
```
mask = d["PETotal_Actl"].eq(0) & d["PETotal_Rtrn"].isna()
d["PETotal_Actl_Corrected"] = d["PETotal_Actl"].mask(mask)
```

5. Results: reported performance

Measure	PE	Public equity
Annual return N	2,991	4,769
Median / mean annual return	11.12% / 11.47%	12.61% / 8.81%
Reported annual standard deviation	15.48%	16.70%
Matched PE minus equity: median / mean	1.04% / 2.06%	N=2,961 paired plan-years



Restyled chart: annual medians from different reporting samples. PE N ranges 57-167; equity N 127-218 by year. The lines show common cycles but do not represent matched portfolios.



Restyled chart: allocation versus return, N=2,991 complete PE plan-years. Pooled r=0.088; the cloud shows only a weak positive linear association, not a benefit from allocating more.

Separate-sample pooled medians favor public equity; matched annual PE-minus-equity returns have a positive median of 1.04% (N=2,961). These answer different questions: the median of paired differences need not equal the difference of separate medians. Neither establishes superior risk-adjusted performance.

6. Additional robustness results

Check	Original	Robustness result / N
PE allocation median	5.10%	8.49%; N=3,576 vs 5,061
Public-equity allocation median	52.40%	52.48%; N=5,052 vs 5,061
PE mean / standard deviation	11.47% / 15.48%	11.39% / 14.81%; N=2,991
Equity mean / standard deviation	8.81% / 16.70%	8.83% / 16.47%; N=4,769
PE/equity correlation	0.526	Year-demeaned 0.261; N=2,961
Allocation/PE correlation	0.088	Year-demeaned 0.103; N=2,991

All 1,485 zero PE allocations lack returns; recoding reduces lowest-band allocation N from 2,523 to 1,038. The band's return N remains 690; all return-band statistics and allocation/return r=0.088 (N=2,991) are unchanged. The higher allocation median is conditional on retained allocation records, not an all-plan estimate.

PE clipping limits are -29.05% and 57.33% (60 values changed; N=2,991); equity limits are -34.16% and 43.34% (96 changed; N=4,769). Medians remain 11.12% and 12.61%, respectively. Tail sensitivity is limited for these pooled summaries.

Equity median paired with PE median	All available: r (N years)	Common years: r (N years)
Current year	0.769 (25)	0.718 (23)
Lag 1 fiscal year(s)	0.341 (24)	0.268 (23)
Lag 2 fiscal year(s)	-0.246 (23)	-0.246 (23)

The contemporaneous median association is strongest, including on common years. This lag check does not support a dominant delayed-equity explanation, but cannot identify or rule out appraisal smoothing. Annual-median samples change over time and are short; nominal lag-test p-values in the saved output are exploratory and do not adjust for serial dependence.

Year-effect regression (statsmodels): equity slope 0.275, clustered SE 0.032, 95% CI [0.212, 0.338], N=2,961, 186 plan clusters. A one-percentage-point equity-return difference is associated with 0.275 percentage points of PE return within a fiscal year. Winsorized slope 0.271 (N=2,961); excluding final FY 0.275 (N=2,866). No plan fixed effects are included; this is not causal.

Excluding FY2025 leaves pooled PE/equity r=0.532 (N=2,866), versus 0.526 (N=2,961). Incomplete final-year coverage does not materially change this association.

Code excerpt | Pair-complete, within-year return correlation.

```
paired = d.dropna(subset=["PETotal_Rtrn", "EQTotal_Rtrn"])
returns = paired[["PETotal_Rtrn", "EQTotal_Rtrn"]]
within_year = returns - returns.groupby(paired["fy"]).transform("mean")
r = within_year["PETotal_Rtrn"].corr(within_year["EQTotal_Rtrn"])
```

7. Limitations and audit findings

Severity	Finding and consequence
HIGH	Source script deletes prior outputs before downloading. It was not run; originals were preserved.
HIGH	Zero-allocation meaning is unverified. Conditional PE allocation N=3,576 excludes 1,485 zeros, changing the estimand.
MEDIUM	Extreme PE returns: 3 of N=2,991; real estate: 1 of N=3,425. They require upstream review, not automatic deletion.
MEDIUM	FY2025: 134 panel plans, 100 PE and 127 equity returns; prior year: 218, 157, 209. Coverage is partial; return periods are not proven partial.
MEDIUM	Unbalanced and changing samples, separate asset samples and repeated observations limit comparisons. Year adjustment weakens the headline correlation.
MEDIUM	Units assumed decimal; numeric coercion is not logged. Live API metadata, response-header handling and extraction vintage are unverified.
LOW	Alternative identifier spellings are discovered but not normalized by the source cleaner. Saved data use the expected identifier.
LOW	Original annual/correlation outputs omit some sample sizes. Added outputs supply them; all numeric results reconcile.

The extreme PE values are -62.80%, -53.70%, 112.02% (N=2,991 PE returns). Their source-document accuracy is unverified. Appraisal smoothing and valuation/reporting lags remain hypotheses, not established mechanisms.

Rolling geometric-return medians are 12.65% (N=2,181 overlapping windows) for PE and 8.30% (N=3,761) for equity. Different samples and overlapping windows preclude interpreting this as an independent or investable matched-return premium.

Fiscal-year labels may represent different year-end dates. Fees, reporting conventions, valuation methods and selection into PE reporting may differ across plans. Plan-clustered errors allow within-plan dependence, not all cross-plan dependence. These records cannot support fund-level IRR, TVPI, DPI, RVPI, cash-flow benchmarking or a causal allocation effect.

8. Possible extensions

Validate extreme observations and zero-allocation codes against public source documentation; capture a dated API snapshot with metadata and non-destructive extraction. Align fiscal year-end dates and gross/net return conventions. Compare matched rolling windows and balanced plan panels; examine within-plan effects, reporting selection and serial-dependence-robust distributed-lag specifications. Fund-level cash-flow analysis would require a separate dataset.

Reproducibility: analysis_extension.py retains original columns and adds corrected/winsorized columns in a separate output panel. Companion CSVs contain sample counts, annual coverage, outliers, lag inputs, model estimates and reconciliation checks. audit_report.txt documents the full issue list and verification boundaries.